1. recall that the materials we use are chemicals or mixtures of chemicals, and state examples;
2. recall that materials can be obtained or made from living things, and give examples such as cotton, paper, silk and wool;
3. understand that there are synthetic materials which are alternatives to materials from living things;
4. interpret representations of rearrangements of atoms during a chemical reaction;
5. understand that during the course of a chemical reaction the numbers of atoms of each element must be the same in the products as in the reactants;
6. recall that crude oil consists mainly of hydrocarbons which are chain molecules of varying lengths made from carbon and hydrogen atoms only;
7. recall that only a small percentage of crude oil is used for chemical synthesis;
8. recall that the petrochemical industry refines crude oil to produce fuels, lubricants and the raw materials for chemical synthesis;
9. understand that some small molecules can join together to make very long molecules called polymers and that the process is called polymerisation;
10. understand that by using polymerisation, a wide range of materials may be produced;
11. recall an example of a material that has replaced an older material because of its superior properties.